

MA2031- Linear Algebra for Engineers

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Module-5 (Lecture note-4)

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Some immediate consequence of spectral theorem

Theorem: Let $A \in M_n(\mathbb{C})$. Then

1. A is self-adjoint $\Leftrightarrow A$ is normal and $\sigma(A) \subseteq \mathbb{R}$.
2. A is unitary $\Leftrightarrow A$ is normal and $\sigma(A) \subseteq \{z \in \mathbb{C} : |z| = 1\}$.
3. A is projection $\Leftrightarrow A$ is normal and $\sigma(A) \subseteq \{0, 1\}$.

Proof (Sketch): Suppose A is normal, hence by spectral theorem

$A = \sum_j \lambda_j P_j$, where λ_j 's are distinct eigenvalues of A and P_j 's are orthogonal projections such that $P_j P_k = 0$ ($j \neq k$) and $\sum_j P_j = I$.

(1) Let $\lambda \in \sigma(A)$ and $0 \neq x \in \mathbb{C}^n$ such that $Ax = \lambda x$. Then

$$\lambda \langle x, x \rangle = \langle \lambda x, x \rangle = \langle Ax, x \rangle = \langle x, A^* x \rangle = \langle x, Ax \rangle = \langle x, \lambda x \rangle = \bar{\lambda} \langle x, x \rangle.$$

Since $\langle x, x \rangle \neq 0$, we get $\lambda = \bar{\lambda}$, so that $\lambda \in \mathbb{R}$. Conversely assume that $\sigma(A) \subseteq \mathbb{R}$. Since $\lambda_i \in \sigma(A) \subseteq \mathbb{R}$, we have $A^* = \sum_j \bar{\lambda}_j P_j^* = \sum_j \lambda_j P_j = A$.

(2) Suppose A is unitary and $\lambda \in \sigma(A)$. Choose $0 \neq x \in \mathbb{C}^n$ such that $Ax = \lambda x$. Then

$$|\lambda|^2 \langle x, x \rangle = \langle \lambda x, \lambda x \rangle = \langle Ax, Ax \rangle = \langle x | A^* Ax \rangle = \langle x | x \rangle,$$

hence $|\lambda| = 1$. Conversely assume that $\sigma(A) \subseteq \{z \in \mathbb{C} : |z| = 1\}$. Then

$$A^* A = \left(\sum_j \lambda_j P_j \right)^* \left(\sum_k \lambda_k P_k \right) = \sum_{j,k} \bar{\lambda}_j \lambda_k P_j P_k = \sum_j |\lambda_j|^2 P_j = \sum_j P_j = I.$$

(3) Exercise.

Some immediate consequence of spectral theorem

Note: Suppose A is normal. Then $A = UDU^*$ for some unitary U and diagonal D . Then

$$A^2 = (UDU^*)(UDU^*) = UD^2U^*$$

$$A^3 = (UD^2U^*)(UDU^*) = UD^3U^*$$

$$A^k = UD^kU^* \quad \forall k \geq 1.$$

Hence, if $p(z) = \sum_{j=0}^k a_j z^j$, then

$$\begin{aligned} p(A) &:= a_0 I + a_1 A + \cdots + a_k A^k = U(a_0 I + a_1 D + \cdots + a_k D^k)U^* \\ &= Up(D)U^*. \end{aligned}$$

Recall: Let $A \in M_n(\mathbb{C})$. Then $A_1 := \frac{A+A^*}{2}$, $A_2 := \frac{A-A^*}{2i}$ are self-adjoint matrices such that

$$A = A_1 + iA_2.$$

Since A_i 's self-adjoint, they are normal. Now from spectral theorem it follows that:

Theorem:

$$M_n(\mathbb{C}) = \text{span}\{\text{self-adjoint matrices}\} = \text{span}\{\text{orthogonal projection matrices}\}.$$

Spectral theorem for real symmetric theorem

Spectral theorem: $A \in M_n(\mathbb{C})$ normal $\Leftrightarrow A = UDU^*$, where $U, D \in M_n(\mathbb{C})$.

Que: Suppose entries of A are real numbers (i.e., $A \in M_n(\mathbb{R})$). Can we choose U, D with real entries?

(Note: $U \in M_n(\mathbb{R})$. Then U unitary $\Leftrightarrow U$ orthogonal i.e., $UU^t = U^tU = I$.)

- ▶ Note that $A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \in M_2(\mathbb{R})$ is normal but does not have (real) eigenvalues.
- ▶ One can show that if $A \in M_n(\mathbb{R})$ is self-adjoint (i.e., $A^t = A$), then A has (real) eigenvalues.

Theorem: Given $A \in M_n(\mathbb{R})$ the following conditions are equivalent:

1. A is symmetric (i.e., $A^t = A$).
2. There exists an orthogonal matrix $U \in M_n(\mathbb{R})$ and diagonal $D \in M_n(\mathbb{R})$ such that $A = UDU^t$.

Exercises

1. Write the following matrices as linear combinations of projection matrices.

$$(a) A = \begin{bmatrix} 1+i & i \\ i & 2+i \end{bmatrix} \quad (b) A = \begin{bmatrix} 1 & 2+i & 1 \\ 2+i & 0 & 1 \\ 1 & 1 & 2+i \end{bmatrix}$$

2. Determine $A^{2019} + A^{2020} + A^{2021}$ where

$$(a) A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \quad (b) A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 2 & 1 & 3 \end{bmatrix}$$